

Web Design

Week 4:JavaScript Basic



JavaScript is the programming language of the web that turns static pages into interactive experiences. While HTML builds the structure and CSS handles the look, JavaScript provides the "brains" and "muscles" powering everything from pop-up menus and live data updates to complex games and mobile apps. Because it runs directly in your browser and has a massive community of developers, it is currently the most popular and versatile language for modern software development.



JavaScript is a true multi-purpose language. While it started as a simple tool for web browsers, it has evolved into one of the most versatile languages in existence. Because it can now run almost anywhere on servers, phones, and even hardware you can use it to build nearly any type of software.

JS Using innerHTML

```
<!DOCTYPE html>
<html>
<body>
<h1>My First Web Page</h1>
<p>My First Paragraph</p>
<p id="demo"></p>
<script>
document.getElementById("demo").innerHTML = "<h2>Hello World</h2>";
</script>
</body>
</html>
```

JS Using Document.Write

```
<!DOCTYPE html>
<html>
<body>
<h1>My First Web Page</h1>
<p>My first paragraph.</p>
<script>
document.write(5 + 6);
</script>
</body>
</html>
```

JS Window alert

```
<!DOCTYPE html>  
<html>  
<body>
```

```
<h1>My First Web Page</h1>  
<p>My first paragraph.</p>
```

```
<script>  
window.alert(5 + 6);  
</script>
```

```
</body>  
</html>
```

JS Console.log

```
<!DOCTYPE html>
```

```
<html>
```

```
<body>
```

```
<script>
```

```
console.log(5 + 6);
```

```
</script>
```

```
</body>
```

```
</html>
```

JS JavaScript Print

```
<!DOCTYPE html>  
<html>  
<body>  
  
<button onclick="window.print()">Print this page</button>  
  
</body>  
</html>
```


JS Variables

```
let name = "Alice";  
const age = 25;  
var city = "New York";
```

```
// can be changed  
// cannot be changed  
// old way (avoid)
```

JS Data Types

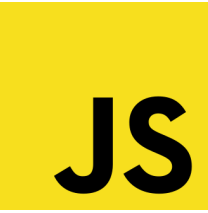
```
let text = "Hello";           // String
let number = 42;              // Number
let isTrue = true;            // Boolean
let nothing = null;           // Null
let notDefined;               // Undefined
```

JS Functions

```
// Regular function
function greet(name) {
  return "Hello, " + name + "!";
}
```

```
// Arrow function (modern)
const greet = (name) => "Hello, " + name + "!";

console.log(greet("Alice")); // Hello, Alice!
```

JS

Conditional Statement

```
let score = 85;
let result = "";
// The logic remains the same
if (score ≥ 90) {
    result = "A";
} else if (score ≥ 80) {
    result = "B";
} else {
    result = "C";
}
// Display the result in the HTML element
document.getElementById("grade-output").innerHTML = result;
```

JS Loop

```
<!DOCTYPE html>
<html>
<body>
  <h3>Counter:</h3>
  <div id="display-loop"></div>

  <script>
    const output = document.getElementById("display-loop");

    for (let i = 0; i < 5; i++) {
      // Appends the number and a comma/space to the div
      output.innerHTML += i + ", ";
    }
  </script>
</body>
</html>
```

**Any Question?
or**

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See you next week!